

EE5606/AI5120 Convex Optimization, Spring 2026

Indian Institute of Technology, Hyderabad

Homework 4: Convex Optimization

Total Marks: **50 points**

Assigned on 19.03.2026, Due date **11:59pm on 26.03.2026**

Instructions

- It is strongly recommended that you work on your homework on an individual basis. If you have any questions or concerns, feel free to talk to the instructor or the TAs.
- Use numpy for basic functions like `log`, `sqrt`, `power`. Do not use other built-in functions.
- Please turn in your work with the following notation for the file name: *your-roll-number-hw4.pdf*.

Question 1

(5 points)

Let $\mathbf{F} \in \mathbb{R}^{n \times m}$, $\hat{\mathbf{x}} \in \mathbb{R}^n$. The restriction of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ to the affine set $\{\mathbf{F}\mathbf{z} + \hat{\mathbf{x}} \mid \mathbf{z} \in \mathbb{R}^m\}$ is defined as the function $\tilde{f} : \mathbb{R}^m \rightarrow \mathbb{R}$ with

$$\tilde{f}(\mathbf{z}) = f(\mathbf{F}\mathbf{z} + \hat{\mathbf{x}}), \quad \text{dom } \tilde{f} = \{\mathbf{z} \mid \mathbf{F}\mathbf{z} + \hat{\mathbf{x}} \in \text{dom } f\}.$$

Suppose f is twice differentiable with a convex domain. Show that $\tilde{f}$ is convex if and only if for all $\mathbf{z} \in \text{dom } \tilde{f}$,

$$\mathbf{F}^T \nabla^2 f(\mathbf{F}\mathbf{z} + \hat{\mathbf{x}}) \mathbf{F} \succeq 0.$$

Question 2

(5+2+3=10 points)

Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a twice differentiable function.

Let X be a positive random variable such that

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2 < \infty.$$

(a) Using Taylor expansion and Jensen's inequality, show that

$$\mathbb{E}[f(X)] - f(\mathbb{E}[X]) \geq \frac{1}{2} f''(\xi) \sigma^2$$

for some ξ between values of X and μ .

Interpret this inequality.

(b) Prove that if f is strictly convex, then

$$\mathbb{E}[f(X)] = f(\mathbb{E}[X]) \iff X \text{ is almost surely constant.}$$



(c) Construct a function f such that:

- Jensen's inequality holds,
- the gap $\mathbb{E}[f(X)] - f(\mathbb{E}[X])$ depends only on μ (not on the distribution of X).

Does such a non-linear function exist? Justify.

Question 3

(5+5 = 10 points)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be defined as

$$f(x) = \log \left(\sum_{i=1}^n e^{x_i} \right)$$

Suppose $p < 1$, $p \neq 0$. and Let $g : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ be defined as

$$g(x) = \left(\sum_{i=1}^n x_i^p \right)^{1/p}.$$

- (a) Show that f is convex.
- (b) Is $g(x)$ Convex or Concave?, Prove it?(Hint:Adapt the proof technique used for part a).

Question 4

(5+5+5 = 15 points)

Show that the following functions are convex.

$f : \mathbb{R}^n \rightarrow \mathbb{R}$

(a)

$$f(x) = \max_{i=1, \dots, k} \|A^{(i)}x - b^{(i)}\|,$$

where $A^{(i)} \in \mathbb{R}^{m \times n}$, $b^{(i)} \in \mathbb{R}^m$, and $\|\cdot\|$ is a norm on $\mathbb{R}^m$.

(b)

$$f(x) = \sum_{i=1}^r |x|_{[i]},$$

where $|x|$ denotes the vector with components $|x_i|$ (i.e., $|x|$ is the absolute value of x componentwise), and $|x|_{[i]}$ is the i th largest component of $|x|$. In other words, $|x|_{[1]}, |x|_{[2]}, \dots, |x|_{[n]}$ are the absolute values of the components of x , sorted in nonincreasing order.

(c)

$$g(x) = -\log \left(-\log \left(\sum_{i=1}^m e^{a_i^T x + b_i} \right) \right)$$

on

$$\text{dom } g = \left\{ x \mid \sum_{i=1}^m e^{a_i^T x + b_i} < 1 \right\}.$$

Question 5

(2+2+6 = 10 points)

Read the paper: Input Convex Neural Networks, Amos & Kolter (2017) Answer the following question.

Consider a neural network defined as:

$$z_{k+1} = \sigma(W_k z_k + U_k x + b_k), \quad k = 0, 1, \dots, L-1,$$

with $z_0 = 0$ and output $f(x) = z_L$, where $x \in \mathbb{R}^n$.

Assume:

- $\sigma(\cdot)$ is convex and non-decreasing
 - $W_k \geq 0$ (element-wise non-negative)
- (a) Explain what it means for the function $f(x)$ to be convex with respect to the input x . Why is convexity an important property in optimization problems involving neural networks?
- (b) Describe the architectural constraints imposed on the network to ensure convexity. In particular, explain the role of non-negative weights and skip connections involving the input x .
- (c) Show that for all k , the function $z_k(x)$ is convex in x . Hence, prove that the final output $f(x)$ is convex.
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